

Banking ClientIQ - Application Wiki

Overview

Banking ClientIQ is an enterprise-grade customer relationship management platform designed for financial institutions. The application provides a comprehensive 360-degree view of banking customers, integrating account data, transaction history, compliance information, and relationship analytics in a unified interface.

Built with modern web technologies (React.js, Node.js, TypeScript) and Material Design principles, the platform emphasizes data density, trust, and operational efficiency for banking professionals.

Core Features

1. Customer Search & Discovery

Intelligent Hybrid Search

- **Multi-criteria search:** Find customers by ID, name, tax ID, government ID, or Silverlake CIF number
- **Fuzzy matching:** Handles typos and spelling variations using PostgreSQL trigram similarity or SQL Server phonetic matching
- **Dual-database support:** Seamlessly works with both PostgreSQL and Microsoft SQL Server backends
- **Relevance scoring:** Returns match confidence scores (0-100%) with exact/fuzzy/partial indicators
- **Real-time results:** Instant search with comprehensive customer matching

Search Examples:

- Customer ID: "3" → 100% exact match
- Fuzzy name: "Smyth" finds "Smith" with 33% similarity
- Tax ID: "123-45-6789" → exact match with full customer details

2. Customer Profile Management

Polymorphic Customer Data Model

- **Individual customers:** First name + last name required
- **Business customers:** Business name required
- **Trust customers:** Business name required
- **Database-enforced validation:** CHECK constraints prevent invalid data combinations
- **Unified search field:** Auto-populated full_name for consistent search experience

Customer Information Display:

- Personal/business details with VIP and Employee badges
- Customer status indicators (Active, Inactive, Closed)

- Birthday notifications with animated indicators
- Date of birth and customer since dates (MM/DD/YYYY format)
- Contact information and address details
- Household relationship mapping

3. Account & Relationship Summary

Total Relationship Summary

- **Deposits overview:** Total deposit balances across all accounts
- **Loans overview:** Total loan balances with separate tracking
- **Independent QoQ analytics:** Separate quarter-over-quarter calculations for deposits and loans
- **Real historical data:** Compares current balances to 90-day historical snapshots
- **Accurate comparisons:** Excludes accounts opened after historical cutoff to prevent inflated metrics

Example Metrics:

- Customer deposits: \$35,078.34 (QoQ: +52.03%)
- Customer loans: \$51,150.80 (QoQ: -13.30%)

4. Interactive Account Management

Account Selection & Filtering

- **Dynamic account list:** View all customer accounts grouped by type
- **Interactive selection:** Click any account to view detailed transaction history
- **CIF number display:** Full Customer Information File (CIF) numbers with search capability
- **Account types supported:**
 - Checking accounts
 - Savings accounts
 - Money Market accounts
 - Certificates of Deposit (CDs)
 - Mortgage loans
 - Auto loans
 - Personal loans
 - Home Equity Lines of Credit (HELOC)
 - Credit cards

Account Details:

- Account numbers and types
- Current balances with real-time accuracy
- Interest rates and terms
- Open dates and maturity dates (for time-based accounts)
- Status indicators

5. Transaction History & Analytics

Comprehensive Transaction Tracking

- **Real transaction data:** Extensive realistic test data with authentic banking patterns
- **Transaction types:** Deposits, withdrawals, transfers, payments, fees, interest
- **Ledger balance tracking:** Accurate running balance after each transaction
- **Date/time display:** All timestamps in PST timezone
- **Null safety:** Comprehensive error handling for missing or incomplete data

Transaction Filtering:

- Filter by selected account
- View account-specific transaction history
- See running ledger balances
- Track transaction patterns over time

6. Balance Trends & Analytics

Real Data-Driven Charts

- **Historical balance trends:** Visualize balance changes over time using actual transaction history
- **Account type breakdown:** Weighted average calculations segmented by account type
- **Interactive charts:** Responsive data visualizations with Material Design aesthetics
- **No synthetic data:** All charts display real transaction-based calculations

7. Debit Card Management

Card Lifecycle Management

- **Card inventory:** View all debit cards linked to customer accounts
- **Status tracking:** Active, Inactive, Expired, Blocked card states
- **Security features:** Last 4 digits display for security
- **Expiration monitoring:** Track card expiration dates
- **Multi-card support:** Manage multiple cards per account

8. Enterprise Notes Module

Collaborative Note-Taking

- **Rich note creation:** Add detailed notes with timestamps
- **Note categorization:** Organize by type (General, Follow-up, Complaint, Compliance)
- **Audit trail:** Track note creation dates and authors
- **Search and filter:** Find notes across customer profiles
- **Compliance support:** Document regulatory interactions and issues

9. Risk & Compliance Management

Regulatory Compliance

- **Risk assessment scores:** Customer risk ratings (Low, Medium, High)
- **KYC status tracking:** Know Your Customer verification status
- **AML monitoring:** Anti-Money Laundering compliance indicators
- **OFAC screening:** Office of Foreign Assets Control check results
- **Last review dates:** Track compliance review schedules
- **Next review scheduling:** Automated compliance calendar

10. Household Relationship Mapping

Family & Business Relationships

- **Relationship types:** Spouse, parent, child, sibling, business partner, authorized signer
- **Cross-customer linking:** Navigate between related customer profiles
- **Household aggregation:** View consolidated household relationships
- **Authority tracking:** Identify authorized signers and co-owners

Technical Capabilities

Architecture

- **Frontend:** React.js with Material-UI and shadcn/ui components
- **Backend:** Express.js with TypeScript
- **Database:** Dual support for PostgreSQL and Microsoft SQL Server
- **ORM:** Drizzle ORM with type-safe database operations
- **State Management:** React Query for server state
- **Routing:** Wouter for client-side navigation

Data Quality & Integrity

- **Null safety:** Comprehensive COALESCE statements and error handling
- **Data validation:** Zod schemas for type-safe API contracts
- **Database constraints:** CHECK constraints enforce business rules at database level
- **Automatic triggers:** PostgreSQL triggers maintain computed fields
- **Computed columns:** SQL Server persisted columns for performance
- **Index optimization:** Trigram GIN indexes and Full-Text Search support

Dual-Database Support

- **Database-agnostic architecture:** SearchProvider adapter pattern isolates vendor-specific logic
- **Automatic detection:** Runtime database vendor detection via environment variables
- **Feature parity:** Identical fuzzy search capabilities on both platforms
- **Vendor-specific migrations:** Separate migration directories for each database
- **Consistent API:** Same response format regardless of backend database

CI/CD Governance

- **Pre-commit validation:** Block commits with governance violations
- **Automated testing:** TypeScript validation, schema consistency checks
- **Contract validation:** API DTO compliance enforcement
- **Security scanning:** npm audit integration
- **Architecture standards:** Automated enforcement of coding patterns

Design System

Material Design Implementation

- **Color palette:**
 - Primary: #2e7d32 (Professional green)
 - Secondary: #936b06 (Gold accents for VIP/premium features)
- **Typography:** Roboto font family (regular, medium, mono)
- **Spacing:** Consistent 8px grid system
- **Components:** Material-UI cards, data tables, and form controls
- **Theme:** Professional banking aesthetic emphasizing trust and stability

User Experience Features

- **Dense data tables:** Maximize information visibility
- **Visual badges:** VIP, Employee, Birthday indicators
- **Responsive layout:** Tabbed navigation for different data sections
- **Date formatting:** Consistent MM/DD/YYYY format throughout
- **PST timezone:** All dates/times displayed in Pacific Standard Time
- **Loading states:** Skeleton screens and progress indicators

Data Model Highlights

Jack Henry Banking Integration

The application is designed to integrate with Jack Henry banking core systems, following industry-standard data models for:

- Customer Information File (CIF) numbers
- Silverlake account structures
- Standard banking transaction codes
- Regulatory compliance fields

Test Data

- **505 total customers:** Mix of individual, business, and trust customers
- **175 individual customers:** Personal banking customers
- **159 business customers:** Commercial banking clients

- **171 trust customers:** Estate and trust accounts
- **Extensive transaction history:** Realistic banking patterns across 90+ days
- **Account diversity:** Multiple account types per customer with varied balances

Key Differentiators

1. **Separate QoQ Calculations:** Deposits and loans show independent quarter-over-quarter growth rates
2. **Historical Accuracy:** Excludes newly-opened accounts from historical comparisons
3. **Real Balance Aggregation:** Uses ledger_balance_after from actual transactions, falling back to account.balance
4. **Polymorphic Customers:** Database-enforced conditional name requirements by customer type
5. **Dual-Database Architecture:** Seamlessly switch between PostgreSQL and SQL Server
6. **Enterprise Governance:** CI/CD framework prevents architectural violations
7. **Comprehensive Null Safety:** Defensive programming throughout the stack

Future Enhancements

Based on architect recommendations:

- Automated regression tests for accounts with/without transaction history
- Shared balance aggregation helper to prevent endpoint divergence
- Expanded analytics dashboards with additional trending metrics
- Enhanced household aggregation views
- Additional compliance workflow automation